



Post

Global Birth Rates Are Probably Lower than You Think

By Patrick Norrick

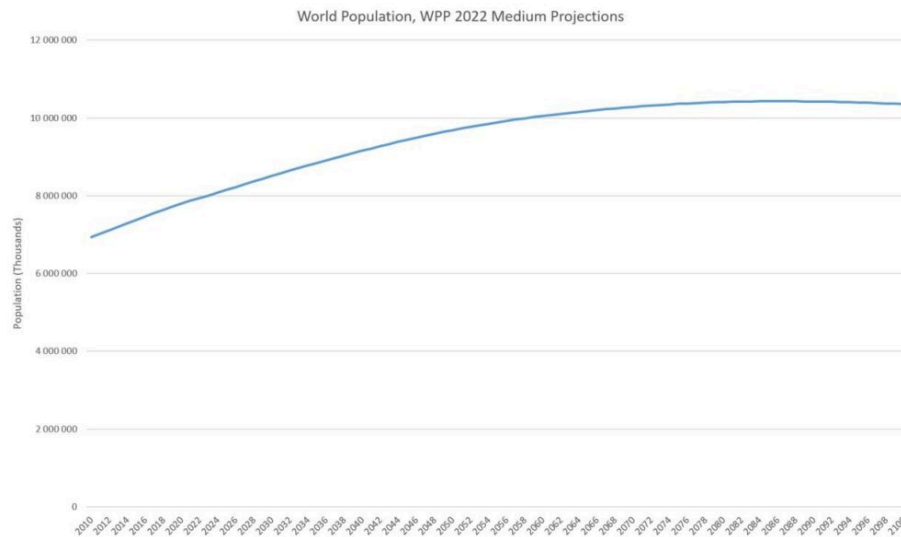
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May 30, 2024

[Global fertility rates are collapsing](#). Faster than many experts predicted even a few years ago, and faster, I believe, than many realize today. The most cited and trusted source for global fertility projections is the United Nations Population Division's [World Population Prospects](#) (WPP). Released approximately every two years, WPP projects birth rates for individual countries and the world as a whole out to the year 2100. Their "medium variant" projections are used as the primary source of fertility data for many researchers, policymakers, and journalists.

Under the most recent projections, released in July of 2022, the projected global fertility rate in 2023 was 2.31, just over the "[replacement level](#)" of fertility required to keep the global population constant in the long run (which, because of selective female abortions and high mortality rates in many parts of the world, is actually higher than the much-discussed benchmark for rich countries of 2.1 births per woman per lifetime). By the UN's reckoning, global fertility will fall below

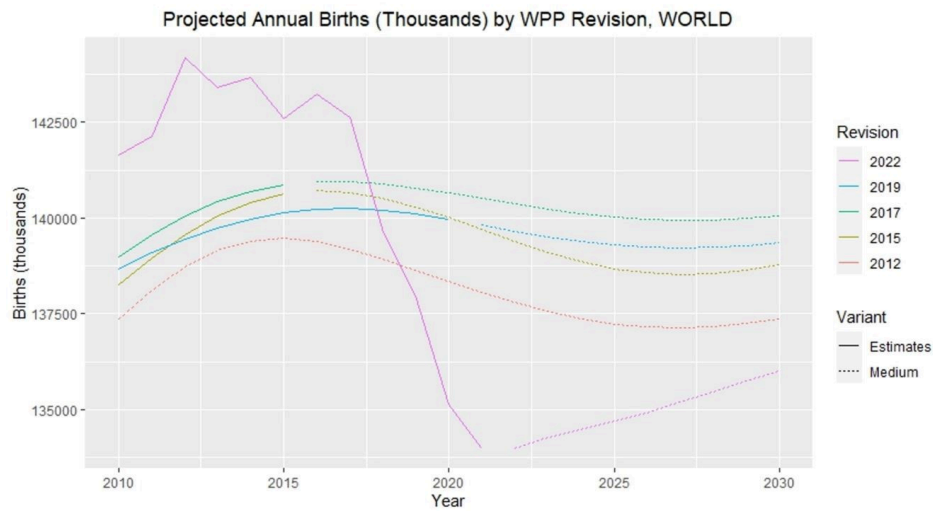
the global replacement level of 2.2-2.3 sometime in the 2040s or 2050s, and the 2022 revision of WPP projects the global population to peak in 2086 at 10.43 billion and then decline thereafter.



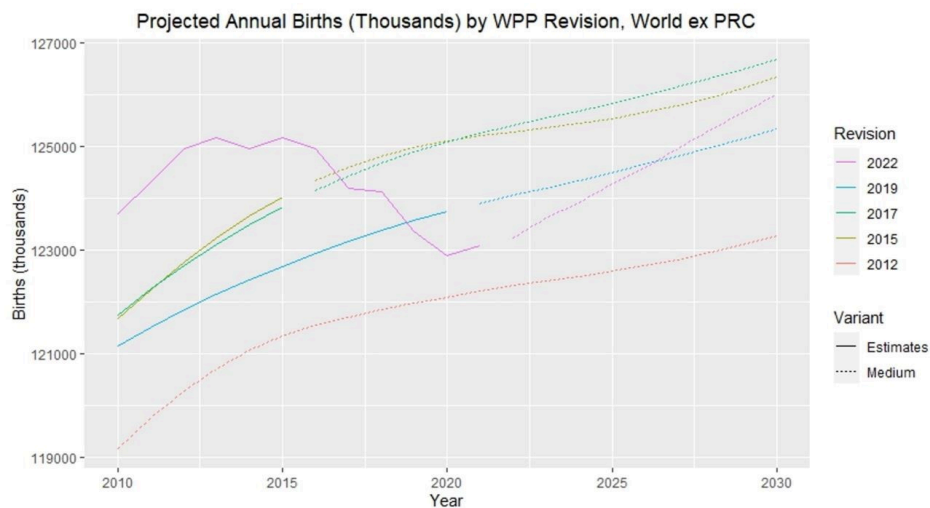
United Nations Department of Economic and Social Affairs, Population Division, "World Population Prospects 2022", <https://population.un.org/wpp/>

Because sustained global depopulation has **never occurred in recorded human history**, it's important to begin to think through the effects and implications of global sub-replacement fertility. It is equally important, however, to get a good estimate of when this may take place. Whether policymakers have six decades or six years before global depopulation begins may make a big difference in their ability to design sensible policies and **adapt to this new world**.

So how quickly is fertility falling according to the UN? The first thing to realize is that projections, and even estimates of past births, often vary considerably from one revision of WPP to the next. Consider, for example, the UN's projections for global births from 2010-2030 from the past several revisions (released, respectively, in 2012, 2015, 2017, 2019, and 2022): there was a large revision in 2022 to reflect the fact that global births were falling much more steeply than the UN had predicted in earlier iterations.



Even if we remove China, the source of much of the variability in their predictions, WPP 2022 still shows large revisions from past versions of their forecasts.

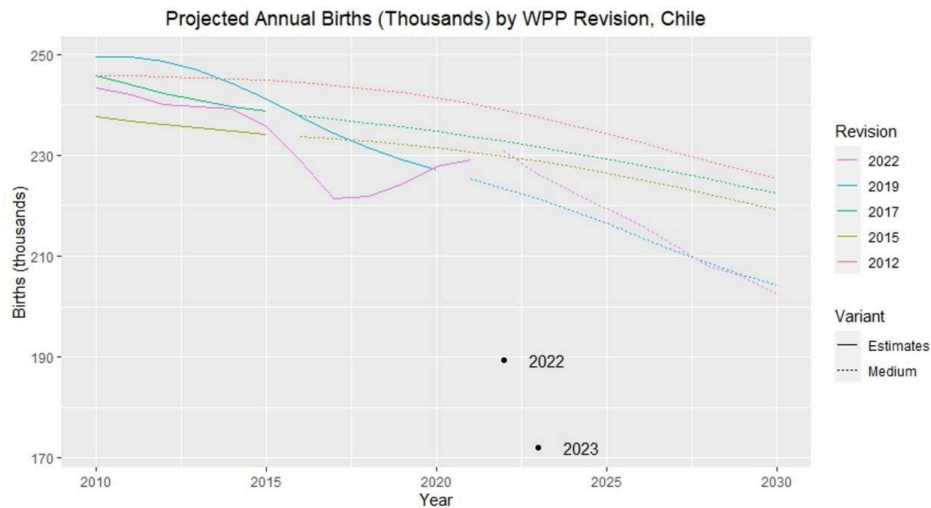


Of particular note is that although the UN has consistently over-predicted birth rates since 2015, their current forecasts show fertility levels recovering from their current trough and rising over the remainder of the decade. If today's low rates of fertility are a hangover effect from the Covid-era, when parents may have altered the timing of when they had children but not the total amount of children they would ultimately have, imagining that births will rebound is reasonable enough. But if fertility rates are falling for underlying, more fundamental reasons, expecting fertility to rebound may be a tall order.

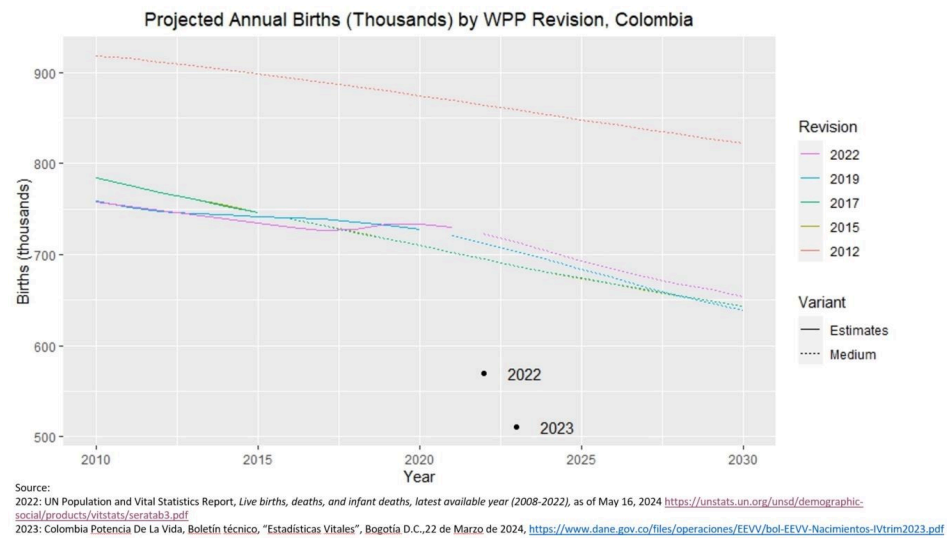
So how have the 2022 revisions of WPP stacked up against the data? How should we expect July's "2024 revision" to change the outlook for global fertility and depopulation? We do not yet have good estimates for worldwide birth levels in 2022 or 2023, but the countries we do have preliminary data for do not paint a pretty picture. I argue that we should expect to see significant downward revisions to the fertility forecasts in the 2024 revision of WPP, possibly, as University of Pennsylvania Economist Jesús Fernández-Villaverde [has argued](#), even going so far as to say that the global fertility rate is *already* below the replacement level. And if not, we are probably not more than a few years away.

Using preliminary reports from 2022 and 2023, births have continued to fall much more rapidly than anticipated in, for example,

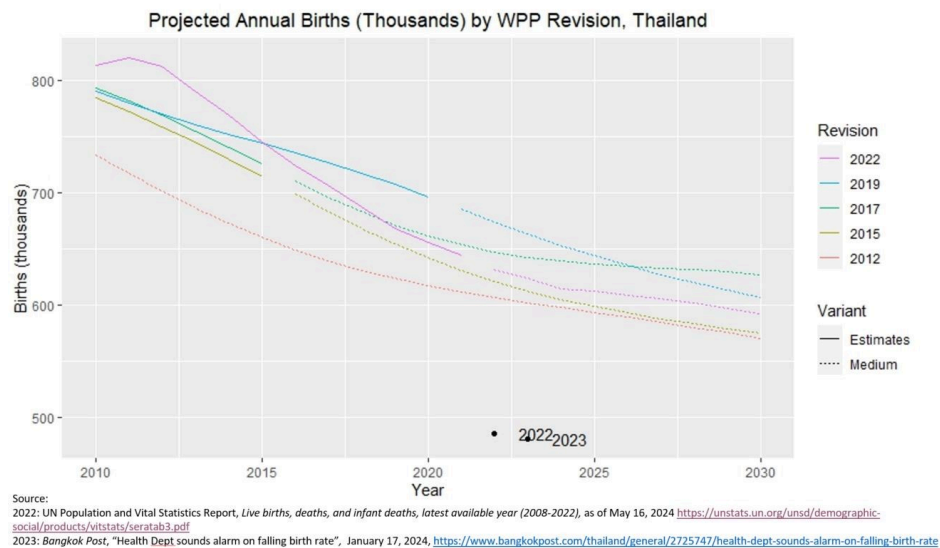
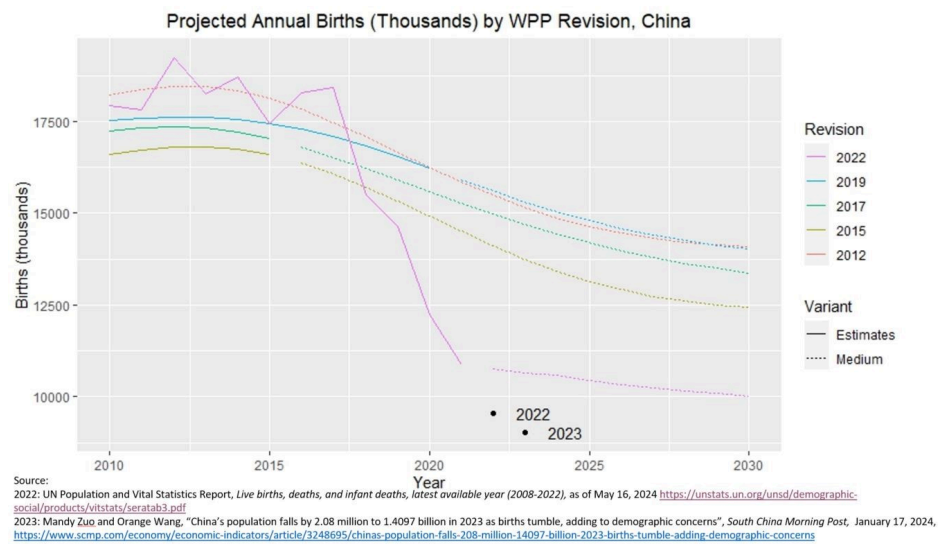
Latin America:



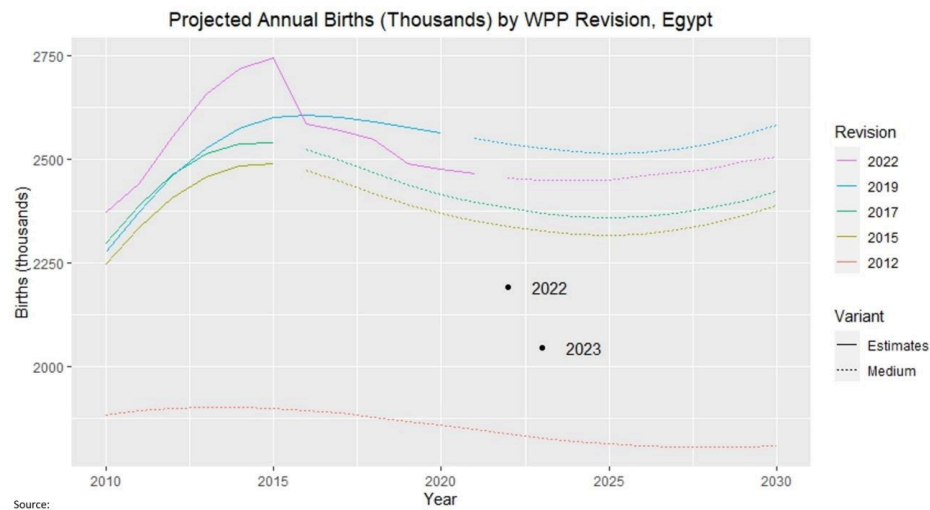
Source
2022 & 2023: *Instituto Nacional de Estadísticas, Chile, "Boletín Demográfico Anual Provisional De Estadísticas Vitales 2023", Marzo 2024, https://www.inec.cl/docs/default-source/nacimientos-matrimonios-y-defunciones/publicaciones-y-anuarios/anuarios-de-estadisticas-vitales/estadisticas-vitales-cifras-provisionales-2023.pdf?sfvrsn=805c0479_9*



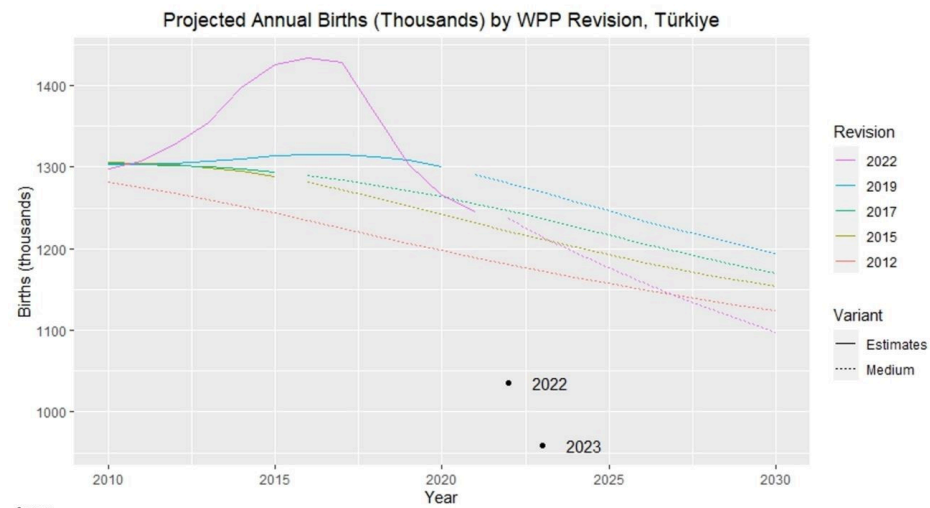
Asia:



Africa and the Middle East:

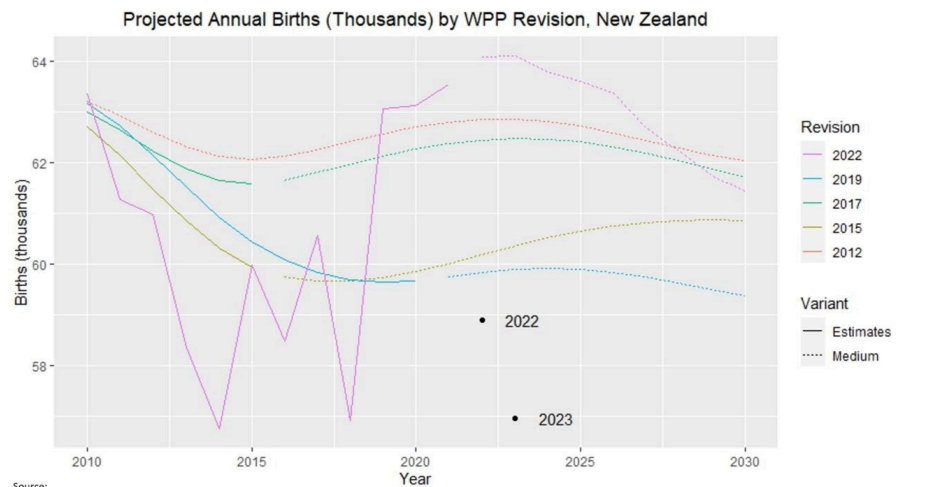


Source:
 2022: UN Population and Vital Statistics Report, *Live births, deaths, and infant deaths, latest available year (2008-2022)*, as of May 16, 2024 <https://unstats.un.org/unsd/demographic-social/products/vitstats/seratab3.pdf>
 2023: Yara Abi Farraj, "Egypt's population declines by 8 percent in 2023", *Economy Middle East*, January 4, 2024, <https://economymiddleeast.com/news/egypt-global-population-rates/>



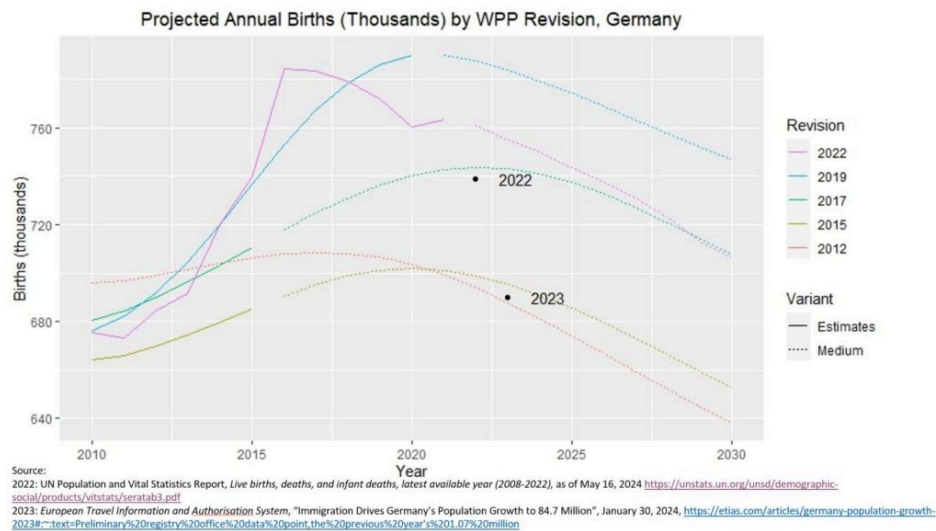
Source:
 2022: UN Population and Vital Statistics Report, *Live births, deaths, and infant deaths, latest available year (2008-2022)*, as of May 16, 2024 <https://unstats.un.org/unsd/demographic-social/products/vitstats/seratab3.pdf>
 2023: Türkiye İstatistik Kurumu, "Doğum İstatistikleri, 2023", 15 Mayıs 2024, <https://data.tuik.gov.tr/Bulten/index?p=Birth-Statistics-2023-53708>

Oceania:

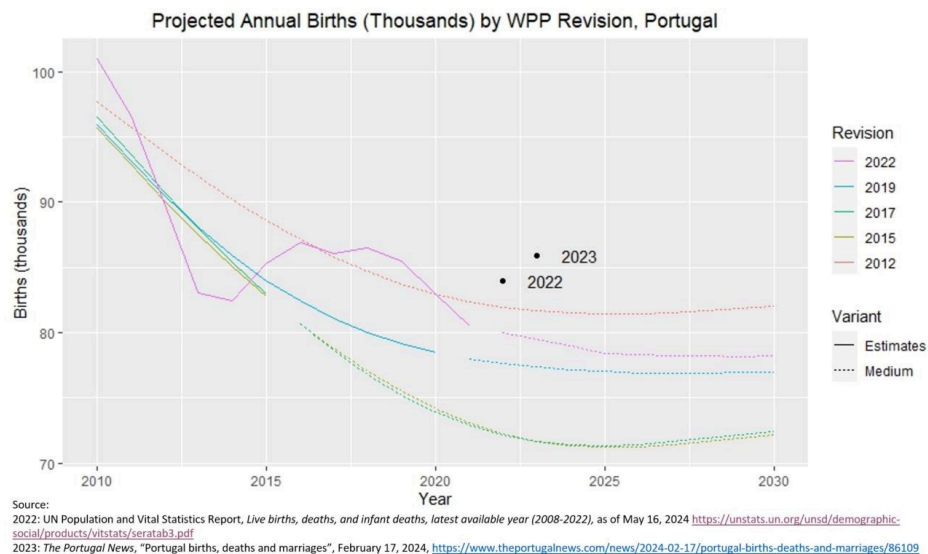


Source:
 2022: UN Population and Vital Statistics Report, *Live births, deaths, and infant deaths, latest available year (2008-2022)*, as of May 16, 2024 <https://unstats.un.org/unsd/demographic-social/products/vitstats/seratab3.pdf>
 2023: Stats NZ, *Tātauranga Aotearoa, "Births and deaths: Year ended December 2023 (including abridged period life table)"*, February 19, 2024, <https://www.stats.govt.nz/information-releases/births-and-deaths-year-ended-december-2023-including-abridged-period-life-table/#:~:text=Births%20in%202023%20dropped%20to,been%20below%2060%20000%20since%202016.>

Europe:



Admittedly, though this list of countries where the UN overestimated births is nowhere near exhaustive, there are a few cases in which the UN underestimated births:



It is also important to remember that all of these birth figures are *estimates* and subject to change. We do not, for example, have any reliable birth data from most countries in Sub-Saharan Africa, where the fertility rate is likely still above 3.0. Nonetheless, for the non-OECD countries with relatively accurate vital registration systems, I estimate that that UN over predicted births in 2022 and 2023 by anywhere from 5-20 percent. This significant dent in the global total fertility rate means that the point at which the world goes sub-replacement and the year depopulation commences are

likely much closer than what the UN predicted in 2022. Will they adjust accordingly in 2024?

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